1. ConvertAudioEngine-DDP51.ps1 *audio-processing script*

Downmixes AAC 7.1, EAC3 7.1, TrueHD 7.1, PCM/FLAC 7.1 to DDP 5.1 (1024k)

- Re-encodes **5.1 audio** to **DDP 5.1 (768k)**
- Encodes stereo audio to DDP 2.0 (256k or 384k depending on codec family)
- Passthrough for **EAC3 Atmos**, **EAC3 5.1**, **EAC3 2.0**, and **TrueHD 5.1**
- Maintains perfect sync on long-duration files
- Normalizes malformed channel layouts
- Removes 2-channel commentary tracks

Requirements

- **PowerShell 7.6** or newer (strict requirement)
- **FFmpeg 8.1 or newer** — (ffmpeg and ffprobe must be placed in the same folder as the script.)

You can grab the latest builds from the official FFmpeg download page:
https://ffmpeg.org/download.html

Quick Start (Windows 11)

1. Place `ConvertAudioEngine-DDP51.ps1`, `ffmpeg.exe`, and `ffprobe.exe` in the same folder.
2. Run the script:

```
pwsh -ExecutionPolicy Bypass -File .\ConvertAudioEngine-DDP51.ps1 ".\YourMovie.mkv"
```

ℹ️ The -ExecutionPolicy Bypass only applies to this single run and doesn't change your system settings.

macOS / Linux

1. Place the script, `ffmpeg`, and `ffprobe` in the same folder.
2. Make the binaries executable (first time only):

```
chmod +x ffmpeg ffprobe
```

1. Run:

```
pwsh -File ./ConvertAudioEngine-DDP51.ps1 "./YourMovie.mkv"
```

🎧 Supported Audio Codecs

The engine supports and processes the following codec families:

Codec Family	Formats Included
AAC	AAC LC, HE-AAC, AAC 5.1, AAC 7.1
EAC3 / Atmos	EAC3, EAC3-JOC (Atmos)
TrueHD	TrueHD 5.1, TrueHD 7.1, TrueHD+Atmos
DTS	DTS Core, DTS-HD MA, DTS-HD HRA
PCM	pcm*s16le, pcm*s24le, pcm*f32le, pcm*f32be
FLAC	All FLAC channel layouts

2. FFmpeg Global Options:

These are global flags to ensure FFmpeg handles large, high-resolution files without data corruption or sync drift.

- `-threads 8` * User-adjustable (4–16), 0 = automatic
    - **What:** Sets the number of CPU threads the encoder is allowed to use.
        - **Why:** Helps balance encoding speed and system stability. It prevents CPU saturation that can lead to frame drops or system instability during the muxing process.
- `-analyzeduration 200M & -probesize 200M`
    - **What:** Increases the initial data analysis buffer to 200 Megabytes.
        - **Why:** High-bitrate audio like TrueHD with Atmos metadata or AAC 7.1 requires a deep probe to fully parse the channel layout. Standard 5MB, 10MB, 20MB probes often misidentify these tracks, causing the encode to fail or use the wrong channel map.
- `-err_detect ignore_err`
    - **What:** Instructs FFmpeg to bypass minor bitstream errors.
        - **Why:** Prevents a single corrupted audio frame (common in digital rips) from aborting a multi-hour conversion process.
- `-drc_scale 0`
    - **What:** Disables Dynamic Range Compression (DRC).
        - **Why:** Ensures the output audio maintains the full dynamic range of the source. It prevents the encoder from "flattening" the sound or lowering the volume of loud peaks.

2.5 ScanThrottle

Limits how many peak-scan tasks run at the same time.

`$ScanThrottle = [Math]::Max(1, [int]([Environment]::ProcessorCount / $ThreadCount))`

- **Recommended values:** 1 to 4

What ScanThrottle affects

- Only affects **peak scanning**, not encoding
- Prevents CPU overload during parallel scanning (Safe Peak Normalizer)
- Keeps your system responsive while scanning multiple tracks

3. The "De-Sync" Safeguard

These flags are applied after the input to maintain perfect alignment on long-duration files.

- `-avoid_negative_ts make_zero`
  - **What:** Forces all stream timestamps to start at absolute zero.
    - **Why:** Many digital files contain negative initial "Presentation Time Stamps" (PTS); this clamps them to zero to prevent audio lag or lead.
- `-max_muxing_queue_size 14000` *(1024-100000) recommended range
  - **What:** Increases the muxing buffer headroom.
    - **Why:** Audio down-mixing and re-encoding (AAC 7.1, TrueHD 7.1, ...) can run slower than the muxer, causing FFmpeg's queue to overflow and trigger errors. 14000 provides enough headroom to prevent these overflows on **all movie lengths.**
- `-map 0:v? -c:v copy`
  - **What:** Maps all video tracks and sets them to stream-copy mode.
    - **Why:** Ensures zero quality loss for the video while moving at maximum speed.
- `-map_metadata 0 -map_chapters 0`
  - **What:** Copies global metadata and chapter markers from the source.
    - **Why:** Preserves the original file's organizational structure (Title, Year, Scene markers).

3.2 Safe Peak Normalizer (SPN)

- **What:** Safe Peak Normalizer checks each audio track before processing and only steps in when the volume is too close to clipping (above –0.5 dBFS).
- **Why:** Some movies arrive with peaks already above 0 dBFS. When that happens, even simple mixing of the channels can push those peaks higher and cause distortion in the final encode. SPN catches this early so the issue never reaches the rest of the audio processing.
- **How:** Safe Peak Normalizer runs right after the script decides each track's action (Encode, Downmix, etc.). If the peak scan shows the track is too hot, a light loudness-correction step is added at the very start of that track's filter chain.
- **Clean Sources:** Nothing is changed — the original loudness and dynamics stay exactly the same.
- **Passthrough tracks:** — Peak detection is skipped entirely. Copied streams cannot have filters applied.
- **Dirty Sources:** SPN fixes unsafe peaks without making the movie sound compressed or "over-normalized."

- **Result:** More consistent, safer audio across all movies, keeping clean sources untouched.

Threshold

These are the values Safe Peak Normalizer uses to decide when to step in.

Setting	Value	Location
`$PeakThresholdDB`	−0.5 dBFS	Global Settings block
`loudnorm` target	−24 LUFS	Set in the audio filter settings
True peak ceiling	−1.5 dBTP	Set in the audio filter settings

-The threshold can be adjusted by changing `$PeakThresholdDB` in the Engine: Global Settings

4. Stereo (2.0) Optimization: Dolby Pro Logic II Compatibility

For stereo tracks, the engine uses specific flags to ensure they playback correctly on both standard speakers and surround sound systems.

- **-dsur_mode 1**
  - **What:** Enables the Dolby Surround Mode flag in the EAC3 bitstream.
    - **Why:** Tells the playback receiver that the 2.0 track contains matrixed surround information (like Pro Logic II), allowing it to properly expand the soundstage.
- **-stereo_rematrixing 1**
  - **What:** Explicitly enables rematrixing during the stereo encode.
    - **Why:** Ensures that the left and right channels are balanced correctly during the conversion from higher-channel sources to stereo.

5. Encoding & Downmix Execution

These parameters determine how the audio is shaped and processed during encoding.

- **`-filter:a:$i $panFilter`**
  - **What:** Applies the custom channel-mapping matrix (the Pan Filter).
    - **Why:** Necessary for downmixing 7.1 or 5.1 to smaller layouts. It ensures that "lost" channels (like back-surrounds) are projected into the remaining speakers rather than simply deleted.
- **`-c:a:$i eac3`**
  - **What:** Sets the codec for audio stream `$i` to Dolby Digital Plus (EAC3).
    - **Why:** Chosen for its high efficiency and broad compatibility with modern smart TVs and home theater receivers.
- **`-b:a:$i $t.Bitrate`**
  - **What:** Sets the target bitrate (e.g., 1024k for 5.1, 256k for 2.0).

        - **Why:** High bitrates ensure the "transparency" of the encode, making it difficult to distinguish from the lossless original.
- `-dialnorm -31`
    - **What:** Sets Dialogue Normalization to the reference "off" position.
        - **Why:** Prevents the playback device from applying its own internal volume leveling, preserving the original mix's intended loudness.
- `-cutoff 20000`
    - **What:** Sets the low-pass filter frequency to 20kHz.
        - **Why:** Preserves the high-frequency "clarity" of the audio that standard encoders sometimes discard to save space.
        - **Applies to:** Downmix, 5.1 re-encode. Not applied to the stereo (2.0) encode path.
- `-metadata:s:a:$i "title=..."`
    - **What:** Injects a custom string into the stream title metadata.
        - **Why:** Allows users to see exactly what the track is (e.g., "DD+ 5.1 Downmix") in their media player's audio selection menu.

6. Audio Rules & Priority

The audio-engine uses a **Priority Mapping (0-100)** to evaluate tracks based on quality and codec type.

EAC3 / Atmos Rules (Priority 100)

Condition	Action	Output	Notes
Atmos (JOC)	Passthrough	Original EAC3 Atmos	Preserves object metadata.
EAC3 7.1	**Downmix**	DDP 5.1 @ 1024k	Pri 99: Maps 8 channels to 6 using Pan Filter.
EAC3 5.1	Passthrough	Original EAC3 5.1	Pri 98
EAC3 2.0	Passthrough	Original EAC3 2.0	Pri 97

TrueHD Rules

Channels	Action	Output	Priority
5.1	Passthrough	Original	81
7.1	Downmix	DDP 5.1 @ 1024k	80

AAC Rules

Channels	Action	Output	Priority
7.1	Downmix	DDP 5.1 @ 1024k	92
5.1	Encode	DDP 5.1 @ 768k	91
2.0	Encode	DDP 2.0 @ 256k	90

DTS Rules

Variant	Action	Output	Priority
DTS-HD MA/HRA (multichannel)	Encode	DDP 5.1 @ 1024k	73
DTS Core (multichannel)	Encode	DDP 5.1 @ 768k	72
DTS-HD 2.0	Encode	DDP 2.0 @ 384k	71
DTS 2.0	Encode	DDP 2.0 @ 256k	70

PCM / FLAC Rules

Channels	Action	Output	Priority
7.1	Downmix	DDP 5.1 @ 1024k	62
5.1	Encode	DDP 5.1 @ 768k	61
2.0	Encode	DDP 2.0 @ 384k	60

7. Spatial Audio: The Pan Filter Matrix

When 7.1 audio is reduced to 5.1, standard downmixing often loses spatial detail from the side channels.

The Formula

```
aformat=channel_layouts=7.1,pan=5.1|FL=FL|FR=FR|FC=FC|LFE=LFE|BL=BL+0.707*SL|BR=BR+0.707*SR,alimiter=limit=0.948:attack=5:release=50:level=disabled
```

Layout Locking (aformat)

What: Forces the decoder to output a strict 8-channel 7.1 layout *before* the Pan Filter.
Why: Some codecs (TrueHD, AAC, EAC3) may decode into non-standard 7.1 variants.
Locking the layout ensures:

- SL really is the Side Left channel
- BL really is the Back Left channel
- No channel swapping

- No silent corruption of the downmix

The Math (0.707)

What: Applies a −3 dB coefficient to the Side Surrounds (SL/SR).
Why: Prevents the side channels from overpowering the rear soundstage and avoids digital clipping.

Limiter (0.948) — For All 7.1 Codecs

alimiter=limit=0.948:attack=5:release=50:level=disabled Why: All 7.1 codecs can produce true peak overshoots after decoding:

- AAC 7.1 > AAC can create peaks slightly above 0 dB after decoding.
- TrueHD 7.1 > Combining similar channels can briefly push the volume higher.
- DTS-HD MA > Decoder can rebuild peaks that go a bit above the original level.
- PCM/FLAC > Short volume spikes can appear when channels are mixed together.
- EAC3 7.1 > Channels are merged into fewer speakers during downmixing.

Limiter ceiling of 0.948 provides about 0.5 dB of true-peak headroom, preventing: 1.) +0.0x dB overshoots 2.) clipping

Important:

- If the source audio already contains true-peak or inter-sample peaks (“clips”), these will also appear in **analysis** of the downmix.
- The limiter prevents new clipping during downmixing and encoding, but **it does not remove overshoots inherent in the source mix**.

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8. Commentary Filtering

Only 2-channel commentary tracks are removed to save space while preserving primary content.

- **Detection Methods:** Title matching, keyword detection, and case-insensitive search.
- **Keywords:** `commentary, director, producer, writer, cast, behind, bonus, alt, interview`.
- **Why 5.1 Commentary is NOT removed:**
  - It is extremely rare in retail media.
    - It is often hard to tell apart from legitimate "Alternate Mixes" or "Home Theater Optimized" tracks.
    - Removing it carries a high risk of "False Positives" (deleting real movie content).

9. Malformed Layout Guards

The script corrects illegal 7-channel layouts:

- **AAC 7ch** → 6ch, **EAC3 7ch** → 6ch, **TrueHD 7ch** → 8ch, **PCM/FLAC 7ch** → 8ch

⛔ Fallback Rules

If no specific rule matches, the following logic is applied:

Channels	Action	Bitrate
≤2ch	Encode	256k
3–6ch	Encode	768k
>6ch	Downmix	1024k

10. Summary Report

At = Audio processing Summary =, the script generates a detailed table with the following columns:

Column	Description
Index	The original stream number (1, 2, 3,...).
Codec	The source format (e.g., TrueHD, DTS).
Channels	The detected number of channels (e.g., 7.1, 5.1).
Action	**Green**: Passthrough, **Yellow**: Downmix, **Blue**: Encode, **Red**: Removed.
Output	The final codec and bitrate configuration.
Priority	The assigned priority level, ranging from 0 to 100.
Rule	the logical rule that was matched during processing, such as "TrueHD 7.1 Rule"

11. Non-Critical FFmpeg Warnings (Safe to Ignore)

During processing, FFmpeg may display one or more of the following messages.
These warnings are **normal**, and do not affect audio quality, sync, or re-encode.
They come from FFmpeg's internal decoders and subtitle parsers — not from the script!

"quant*step*size larger than huff_lsbs"

- **Cause:** Printed by the TrueHD decoder during decoding (not probe).
- **Impact:** Safe to ignore. Does not affect decoding, channel layout, or the Pass+Copy path.

"Codec AVOption drc_scale ... has not been used for any stream"

- **Cause:** (`-drc_scale`) applies only to AC-3/EAC3 decoders.

FFmpeg prints this when the input track is not AC-3 (e.g., TrueHD, DTS, FLAC).

- **Impact:** Safe to ignore. The script disables DRC intentionally for consistent output.

“(Subtitle: hdmv*pgs*subtitle) unspecified size”

- **Cause:** PGS (Blu-ray) subtitles do not declare width/height in the container.

  FFmpeg warns until the first subtitle packet is decoded.

- **Impact:** Safe to ignore. Subtitles are copied untouched (`-c:s copy`).

“Assuming an incorrectly encoded 7.1 channel layout…instead of 7.1(wide)”

- **Cause:** AAC has two 7.1 layouts. Most encoders use the common (non-spec) layout. FFmpeg prints this when the file does not use the strict 7.1(wide) AAC layout.
- **Impact:** Safe to ignore. The script already corrects and pins the layout using (`aformat=channel_layouts=7.1`)